STA 2002, Summer 2024
Probability and Statistics II

Confidence Intervals

Motivation

- **Data scientists often involved with estimating parameters**
  - Maximum likelihood estimate
  - Method of moments estimate
- **Suppose you have observed 10 samples**
- **You can use sample mean to estimate the true (population) mean**
- **But the estimate can be *close* or very *far* from the true mean**
- **To avoid this, report the estimate for a range of plausible values called Confidence Interval**

Reading: Chapter 7.1, 7.2, 7.3, 7.4 in textbook

Confidence Statements

- **Fortune Teller**



“I believe the answer is 75.6 meters”

- **Scientist**



“I believe the answer is 75.6 meters plus or minus 2.0”

Confidence Level: in a nutshell

- A confidence level specifies a confidence level, 90%, 95%, 99% - measures reliability of the estimator

- Confidence interval [L, U], where L and U both depend on the data (so that L and U are both random variables)

- Reliability means if we repeat the experiments over and over again, 95% of times the interval will cover the true parameter

Confidence Interval

Let θ = unknown population parameter

Definition

Let $L < U$ be two numbers. If $[L, U]$ contain the parameter θ with probability $1 - \alpha$. Then $[L, U]$ is called the two-sided confidence interval with confidence level $1 - \alpha$.

Mathematically

$$\mathrm{P}(\theta \in [L, U]) = 1 - \alpha$$

Important

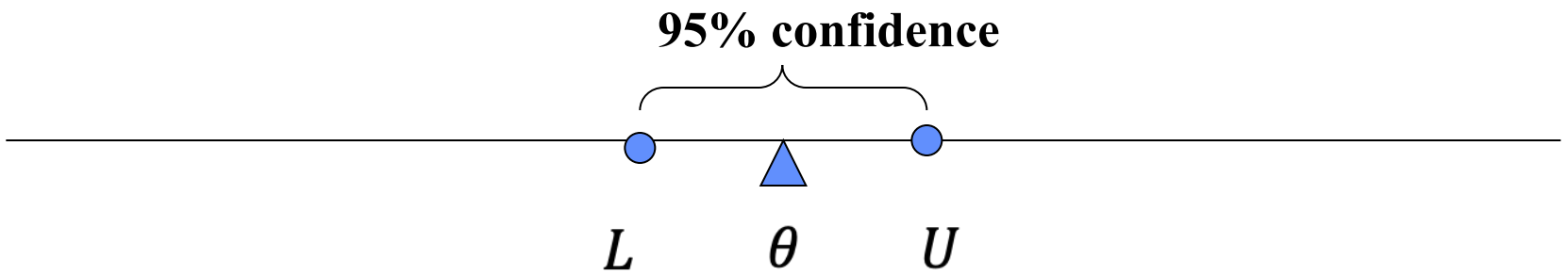
The two sides $L < U$ are constructed from data, and thus random.

True parameter θ is fixed (deterministic) but unknown.

Interpretation

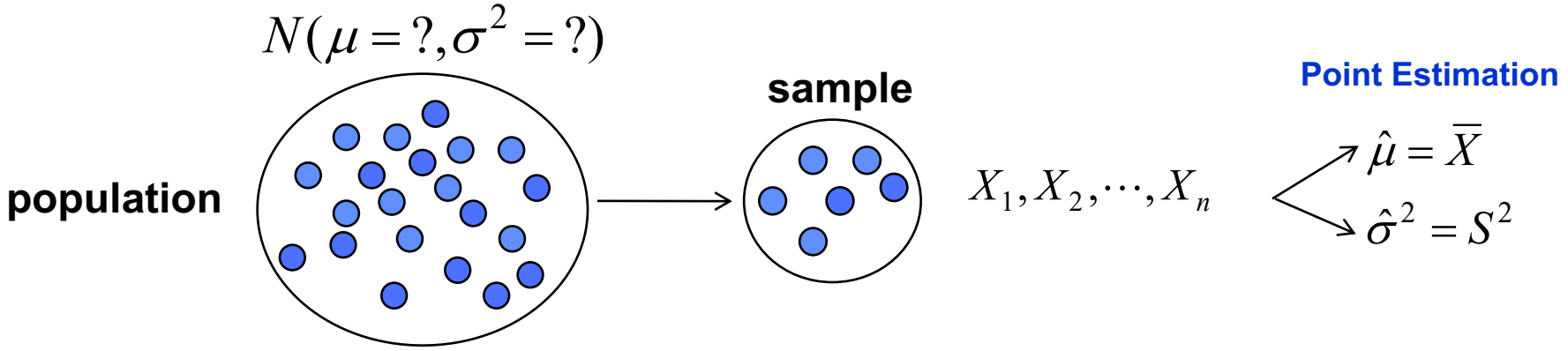
- For example, $\alpha = 0.05$, we expect that 95% of all observed samples would give an interval that includes the true parameter.

$$\mathrm{P}(\theta \in [L, U]) = 1 - \alpha$$



Point Estimation Vs. Confidence Intervals

- The population distribution parameters are unknown. How to estimate the population parameters from samples?
  - Point Estimation / Interval Estimation



Interval Estimation

$$\Pr\{L \le \mu \le U\} = 1 - \alpha \qquad \Pr\{L' \le \sigma^2 \le U'\} = 1 - \alpha$$

1-α : Confidence level

- Both lower and upper bounds are functions of a random sample

$$L = g(X_1, X_2, \cdots, X_n)$$
$$U = h(X_1, X_2, \cdots, X_n)$$

So, how to become a scientist?

- **Fortune Teller**



"I believe the answer is 75.6 meters"

- **Scientist**



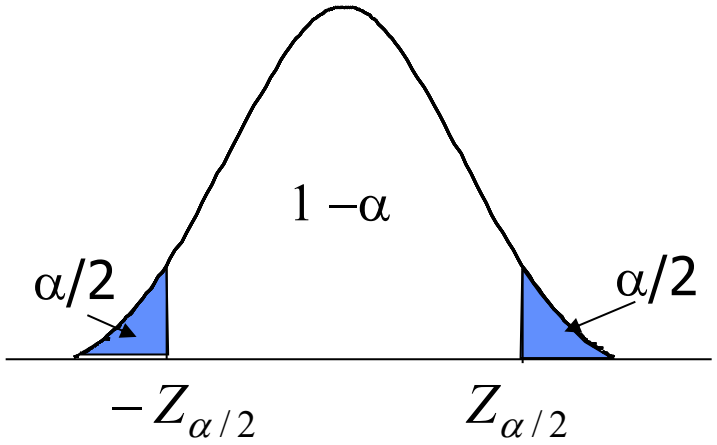
"I believe the answer is 75.6 meters plus or minus 2.0"

Two-sided Confidence Interval (CI) for Mean

100(1-α)% CI for Mean of a Normal Distribution (two-sided, known variance)

- **For mean μ from a normal population with known σ,**

$$X_1, X_2, \cdots, X_n \sim N(\mu, \sigma^2) \rightarrow \bar{X} \sim N(\mu, \frac{\sigma^2}{n}) \quad \Rightarrow \quad \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$



$$P\left\{-z_{\alpha/2} \le \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \le z_{\alpha/2}\right\} = 1 - \alpha$$

Confidence level

CI Lower CI Upper

$$\bar{x} - Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}} \le \mu \le \bar{x} + Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$$

This CI can be used for mean of non-normal distributions when *n*>30 (**CLT**)

Example



You want to rent an unfurnished 1Br apartment for the next semester. The mean monthly rent for a sample of 20 apartments in the local newspaper is $540 for a community where you want to move. Historically, the standard deviation of the rent is $80. Find a 90% confidence interval for the mean monthly rent in this community.

Example: $n = 20$

$\bar{x} = 540$

$\sigma = 80$

$\alpha = 0.1 \quad (1-\alpha = 0.9)$

$\alpha/2 = 0.05$

$Z_{\alpha/2} = Z_{0.05} = 1.645$

$\Rightarrow$ C.I: $540 - 1.645 \times \frac{80}{\sqrt{20}} \leq \mu \leq 540 + 1.645 \times \frac{80}{\sqrt{20}}$

OR. write. $540 \pm 1.645 \times \frac{80}{\sqrt{20}}$: $[510.57, 569.43]$

Example

Example 7.1-1 Let X equal the length of life of a 60-watt light bulb marketed by a certain manufacturer. Assume that the distribution of X is $N(\mu, 1296)$. If a random sample of $n = 27$ bulbs is tested until they burn out, yielding a sample mean of $\bar{x} = 1478$ hours, then a 95% confidence interval for μ is

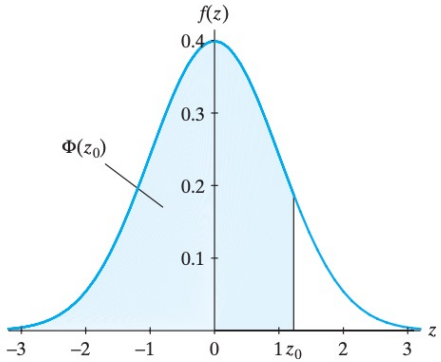
$$\left[\bar{x} - z_{0.025}\left(\frac{\sigma}{\sqrt{n}}\right), \bar{x} + z_{0.025}\left(\frac{\sigma}{\sqrt{n}}\right)\right] = \left[1478 - 1.96\left(\frac{36}{\sqrt{27}}\right), 1478 + 1.96\left(\frac{36}{\sqrt{27}}\right)\right]$$

$$= [1478 - 13.58, 1478 + 13.58]$$

$$= [1464.42, 1491.58].$$

■

Table Va The Standard Normal Distribution Function



$$P(Z \le z) = \Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-w^2/2}\, dw$$

$$\Phi(-z) = 1 - \Phi(z)$$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7703	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

α	0.400	0.300	0.200	0.100	0.050	0.025	0.020	0.010	0.005	0.001
z_α	0.253	0.524	0.842	1.282	1.645	1.960	2.054	2.326	2.576	3.090
$z_{\alpha/2}$	0.842	1.036	1.282	1.645	1.960	2.240	2.326	2.576	2.807	3.291

100(1-α)% CI for Mean of a non-Normal Distribution (two-sided, known variance)

Theorem 5.6-1 **(Central Limit Theorem)** If $\overline{X}$ is the mean of a random sample $X_1, X_2, \ldots, X_n$ of size n from a distribution with a finite mean μ and a finite positive variance σ^2, then the distribution of

$$W = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} = \frac{\sum_{i=1}^{n} X_i - n\mu}{\sqrt{n}\,\sigma}$$

is $N(0,1)$ in the limit as $n \to \infty$.

Using CLT

$$P\left(-z_{\alpha/2} \le \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \le z_{\alpha/2}\right) \approx 1 - \alpha$$

CI Lower CI Upper

$$\bar{x} - Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}} \le \mu \le \bar{x} + Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$$

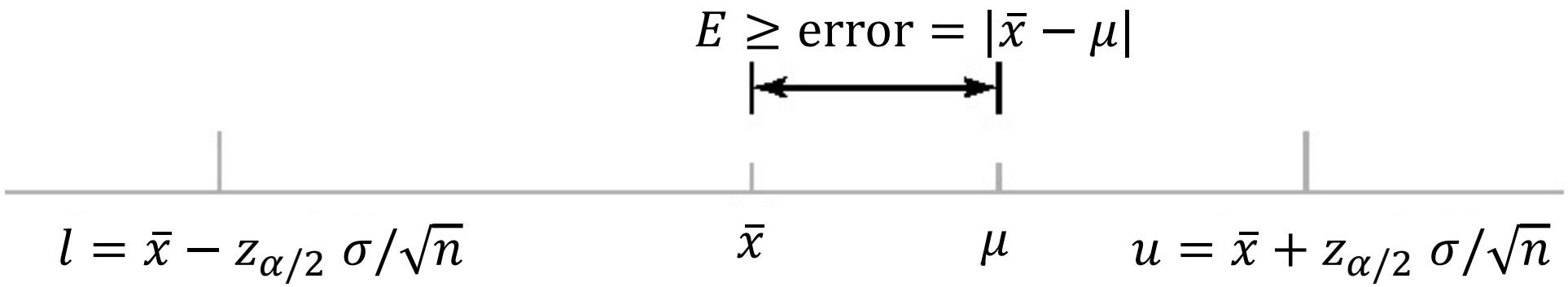
an **approximate** 100(1 − α)% confidence interval for μ

Example

Example 7.1-3 Let X equal the amount of orange juice (in grams per day) consumed by an American. Suppose it is known that the standard deviation of X is $\sigma = 96$. To estimate the mean μ of X, an orange growers' association took a random sample of $n = 576$ Americans and found that they consumed, on the average, $\bar{x} = 133$ grams of orange juice per day. Thus, an approximate 90% confidence interval for μ is

$$133 \pm 1.645\left(\frac{96}{\sqrt{576}}\right), \qquad \text{or} \qquad [133 - 6.58,\ 133 + 6.58] = [126.42,\ 139.58]. \quad \blacksquare$$

Choice of Sample Size

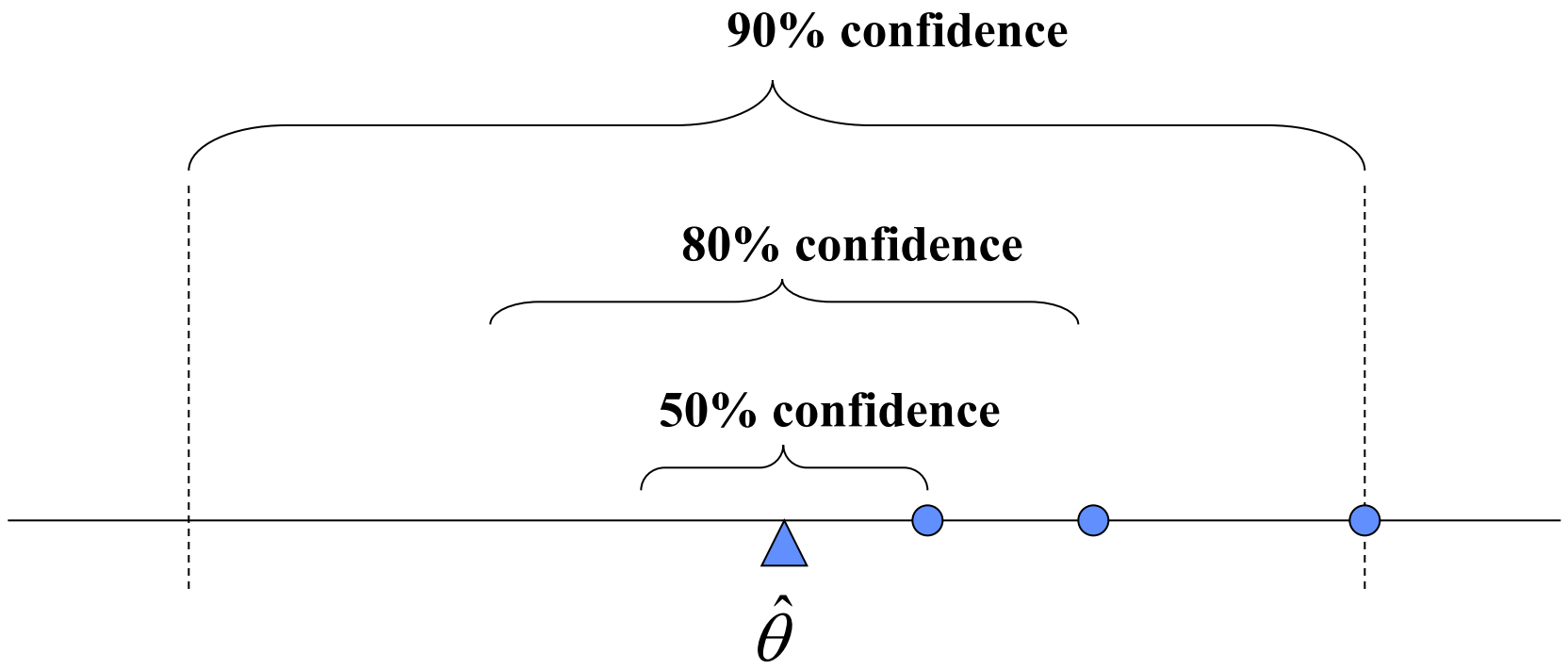


If $\bar{x}$ is used as an estimate of μ, we can be $100(1-\alpha)\%$ confident that the error $|\bar{x} - \mu|$ will not exceed a specified amount E when the sample size is

$$n = \left(\frac{z_{\alpha/2}\sigma}{E}\right)^2$$

Constructing a Confidence Interval

- The bigger the sample size n, the smaller the width of the C.I., resulting in a shorter C.I..
- The bigger the α, the smaller the z_α, resulting in a shorter C.I..



The length of a confidence interval is a measure of the **precision** of estimation.

Choice of Sample Size

A math department wishes to evaluate a new method of teaching calculus with a computer. At the end of the course, the evaluation will be made on the basis of scores of the participating students on a standard test. There is particular interest in **estimating μ, the mean score for students taking the course**. Thus, there is a desire to **determine the number of students, n**, who are to be selected at random from a larger group of students to take the course.

So, let us find the sample size n such that we are fairly confident that $\bar{x} \pm 1$ contains the unknown test mean μ. From past experience, it is believed that the standard deviation associated with this type of test is about 15. Accordingly, using the fact that the sample mean of the test scores, $\bar{x}$, is approximately $N(\mu, \sigma^2/n)$, we see that the interval given by $\bar{x} \pm 1.96(15/\sqrt{n})$ will serve as an approximate 95% confidence interval for μ.

$$1.96\left(\frac{15}{\sqrt{n}}\right) = 1$$

or, equivalently,

$$\sqrt{n} = 29.4, \qquad \text{and thus} \qquad n \approx 864.36,$$

or $n = 865$ because n must be an integer.

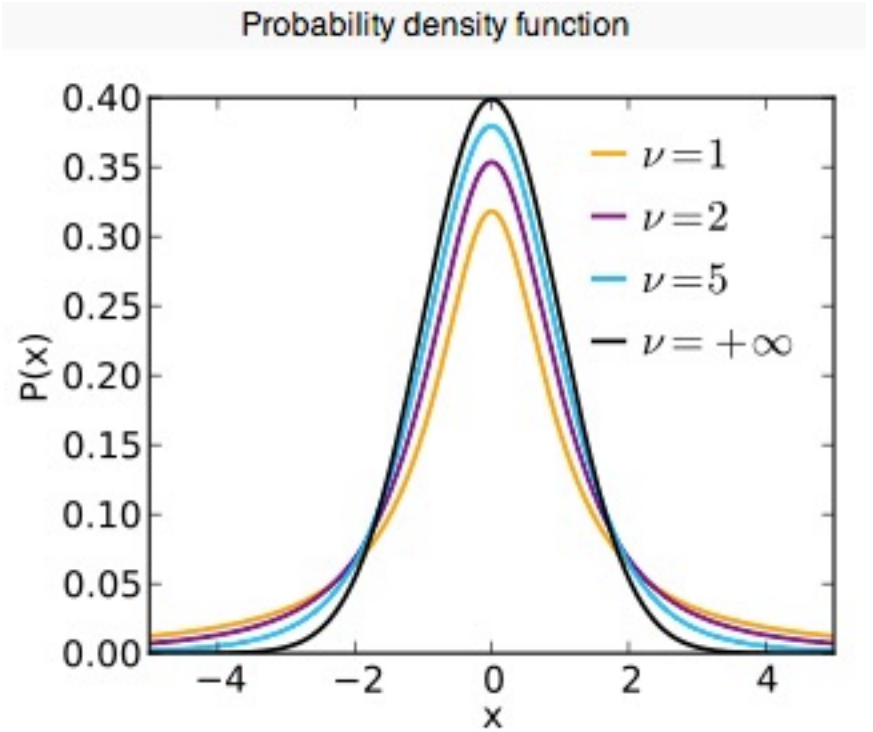
Recall: t-distribution: unknown variance

$$Note: \bar{X} \sim N(\mu, \sigma^2 / n)$$

$$\frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \sim Z$$

$$\frac{\bar{X} - \mu}{S / \sqrt{n}} \sim t(n-1)$$

t distribution

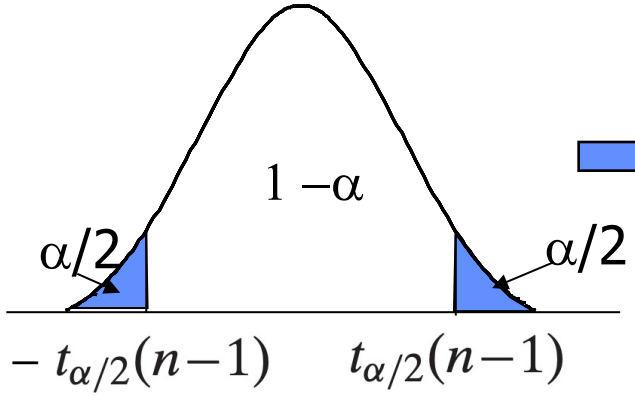


$$\bar{X} = \frac{X_1 + X_2 + \cdots + X_n}{n}$$

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2$$

100(1-α)% CI for Mean of a Normal Distribution (two-sided, unknown variance)

$$X_1, X_2, \cdots, X_n \sim N(\mu, \sigma^2) \rightarrow \frac{\overline{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$$



$$P\left(-t_{\alpha/2}(n-1) \leq \frac{\overline{X} - \mu}{S/\sqrt{n}} \leq t_{\alpha/2}(n-1)\right) = 1 - \alpha$$

$$\overline{X} - t_{\alpha/2}(n-1)\left(\frac{S}{\sqrt{n}}\right) \leq \mu \leq \overline{X} + t_{\alpha/2}(n-1)\left(\frac{S}{\sqrt{n}}\right)$$

Note that $t_{\alpha/2}(n-1) \approx z_{\alpha/2}$ for *n* >= 30

Example



A study of the career path of hotel managers sent questionnaires to random sample of 11. According to these responses, the average time the general managers spent with their current company was 11.7 years and the sample standard deviation of time with the company is 3.2 years. Give a 99% confidence interval for the mean number of years general managers have spent with their current company.

Example. $n = 11$. $\bar{x} = 11.7$. $s = 3.2$

$\alpha = 0.01$ $\quad 1-\alpha = 0.99$ (confidence level)

$\alpha/2 = 0.005$ $\quad t_{\alpha/2}(n-1) = t_{0.005}(10) = 3.169$

$$\Rightarrow \text{C.I.} \quad 11.7 \pm 3.169 \times \frac{3.2}{\sqrt{11}}$$

$$[8.64,\ 14.76]$$

Example

Example 7.1-5

Let X equal the amount of butterfat in pounds produced by a typical cow during a 305-day milk production period between her first and second calves. Assume that the distribution of X is $N(\mu, \sigma^2)$. To estimate μ, a farmer measured the butterfat production for $n = 20$ cows and obtained the following data:

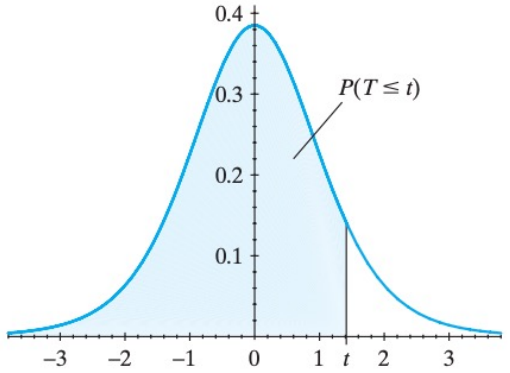
481	537	513	583	453	510	570	500	457	555
618	327	350	643	499	421	505	637	599	392

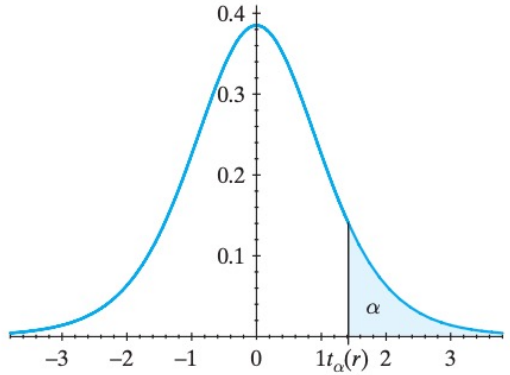
For these data, $\bar{x} = 507.50$ and $s = 89.75$. Thus, a point estimate of μ is $\bar{x} = 507.50$. Since $t_{0.05}(19) = 1.729$, a 90% confidence interval for μ is

$$507.50 \pm 1.729\left(\frac{89.75}{\sqrt{20}}\right) \quad \text{or}$$

$$507.50 \pm 34.70, \quad \text{or equivalently,} \quad [472.80, 542.20].$$

Table VI The t Distribution





$$P(T \le t) = \int_{-\infty}^{t} \frac{\Gamma[(r+1)/2]}{\sqrt{\pi r}\,\Gamma(r/2)(1+w^2/r)^{(r+1)/2}}\,dw$$

$$P(T \le -t) = 1 - P(T \le t)$$

	$P(T \le t)$						
	0.60	0.75	0.90	0.95	0.975	0.99	0.995
r	$t_{0.40}(r)$	$t_{0.25}(r)$	$t_{0.10}(r)$	$t_{0.05}(r)$	$t_{0.025}(r)$	$t_{0.01}(r)$	$t_{0.005}(r)$
1	0.325	1.000	3.078	6.314	12.706	31.821	63.657
2	0.289	0.816	1.886	2.920	4.303	6.965	9.925
3	0.277	0.765	1.638	2.353	3.182	4.541	5.841
4	0.271	0.741	1.533	2.132	2.776	3.747	4.604
5	0.267	0.727	1.476	2.015	2.571	3.365	4.032
6	0.265	0.718	1.440	1.943	2.447	3.143	3.707
7	0.263	0.711	1.415	1.895	2.365	2.998	3.499
8	0.262	0.706	1.397	1.860	2.306	2.896	3.355
9	0.261	0.703	1.383	1.833	2.262	2.821	3.250
10	0.260	0.700	1.372	1.812	2.228	2.764	3.169
11	0.260	0.697	1.363	1.796	2.201	2.718	3.106
12	0.259	0.695	1.356	1.782	2.179	2.681	3.055
13	0.259	0.694	1.350	1.771	2.160	2.650	3.012
14	0.258	0.692	1.345	1.761	2.145	2.624	2.997
15	0.258	0.691	1.341	1.753	2.131	2.602	2.947
16	0.258	0.690	1.337	1.746	2.120	2.583	2.921
17	0.257	0.689	1.333	1.740	2.110	2.567	2.898
18	0.257	0.688	1.330	1.734	2.101	2.552	2.878
19	0.257	0.688	1.328	1.729	2.093	2.539	2.861
20	0.257	0.687	1.325	1.725	2.086	2.528	2.845

100(1-α)% CI for Mean of a non-Normal Distribution (two-sided, unknown variance)

$$\frac{\overline{X}-\mu}{S/\sqrt{n}} \overset{\text{approx}}{\sim} t(n-1)$$

If *n* is large enough (say when *n*>30), or if each X_i is approximately normal, then the approximation works well. Otherwise the method has to be used with caution!

$$\bar{x} - t_{\alpha/2,n-1}s/\sqrt{n} \leq \mu \leq \bar{x} + t_{\alpha/2,n-1}s/\sqrt{n}$$

an **approximate** 100(1 − *α*)% confidence interval for *μ*

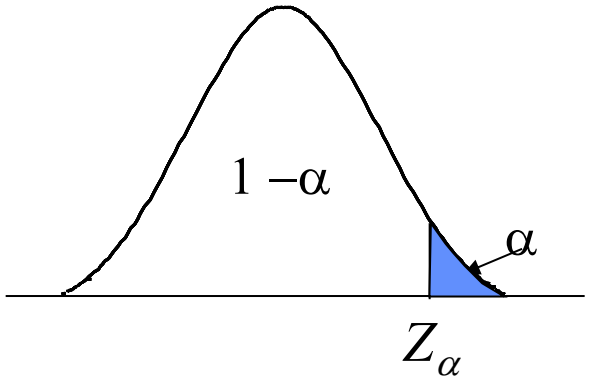
One-sided Confidence Interval (CI) for Mean

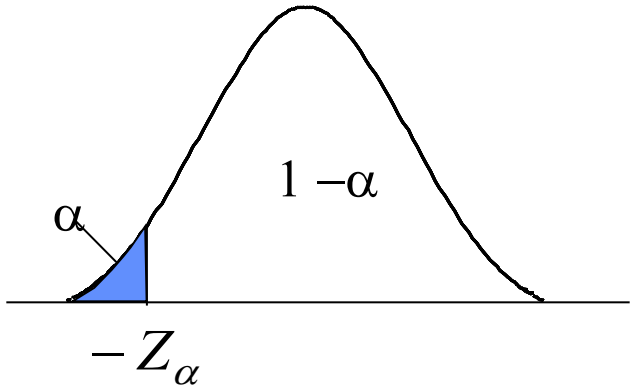
One-sided Confidence Interval

- Confidence interval in the forms of
  - $[-\infty, U]$ or
  - $[L, \infty]$

- We have some prior knowledge about the parameter: it has to be greater than or smaller than certain numbers

100(1-α)% CI for Mean of a Normal Distribution (one-sided, known variance)





$$\Pr\left\{\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \le Z_\alpha\right\} = 1-\alpha$$

$$\Pr\left\{-Z_\alpha \le \frac{\bar{X}-\mu}{\sigma/\sqrt{n}}\right\} = 1-\alpha$$

$$\mu \ge \bar{X} - Z_\alpha \times \frac{\sigma}{\sqrt{n}}$$

Lower bound

$$\mu \le \bar{X} + Z_\alpha \times \frac{\sigma}{\sqrt{n}}$$

Upper bound

This CI can be used for mean of non-normal distributions when n>30

Example

Example: The strength of a disposable plastic beverage container is being investigated. The strengths are normally distributed, with a known standard deviation of 15 psi. A sample of 20 plastic containers has a mean strength of 246 psi. Compute the 95% lower bound CI for the process mean.



Example: $\sigma = 15$, $n = 20$, $\bar{x} = 246$

$\alpha = 0.05$. $1 - \alpha = 0.95$

$z_\alpha = z_{0.05} = 1.645$

lower bound C.I:

$$\mu \geq \bar{x} - z_\alpha \frac{\sigma}{\sqrt{n}} = 246 - 1.645 \times \frac{15}{\sqrt{20}}$$

$$= 240.48$$

$\therefore$ $[240.48, +\infty)$

Upper bound C.I.

$$\mu \leq \bar{x} + z_\alpha \frac{\sigma}{\sqrt{n}} = 246 + 1.645 \times \frac{15}{\sqrt{20}} = 251.52$$

$(-\infty, 251.52]$

100(1-α)% CI for Mean of a Normal Distribution (one-sided, unknown variance)

$$X_1, X_2, \cdots, X_n \sim N(\mu, \sigma^2) \rightarrow \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$$

- **Upper confidence bound for μ:** $\mu \leq \bar{x} + t_\alpha(n-1)(s/\sqrt{n})$

- **Lower confidence bound for μ:** $\mu \geq \bar{x} - t_\alpha(n-1)(s/\sqrt{n})$

Example: Alloy adhesion

An article in the journal *Materials Engineering* (1989, Vol. II, No. 4, pp. 275–281) describes the results of tensile adhesion tests on 22 U-700 alloy specimens. The load at specimen failure is as follows (in megapascals):

19.8	10.1	14.9	7.5	15.4	15.4
15.4	18.5	7.9	12.7	11.9	11.4
11.4	14.1	17.6	16.7	15.8	
19.5	8.8	13.6	11.9	11.4	

The sample mean is $\bar{x} = 13.71$, and the sample standard deviation is $s = 3.55$.

Construct 95% lower confidence interval.



Example: $\bar{x} = 13.71$. $s = 3.55$

$n = 22$

$\alpha = 0.05$ $\quad 1-\alpha = 0.95$

$t_\alpha(n-1) = t_{0.05}(21) = 1.72$

Lower bound C.I

$$\mu \geq \bar{x} - t_\alpha(n-1) \cdot \frac{s}{\sqrt{n}}$$

$$= 13.71 - 1.72 \times \frac{3.55}{\sqrt{22}}$$

$$= 12.41$$

$[12.41, +\infty)$

Upper bound C.I

$$\mu \leq 13.71 + 1.72 \times \frac{3.55}{\sqrt{22}}$$

$(-\infty, 15.01]$

Confidence Interval for Variance

100(1-α)% CI for Variance of a Normal Distribution

- Using chi-square, we can estimate a CI for the variance, σ^2, of a normal population.

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1} \quad \Rightarrow \quad P\left\{\chi^2_{1-\alpha/2}(n-1) \le \frac{(n-1)s^2}{\sigma^2} \le \chi^2_{\alpha/2}(n-1)\right\} = 1-\alpha$$

Two-sided CI

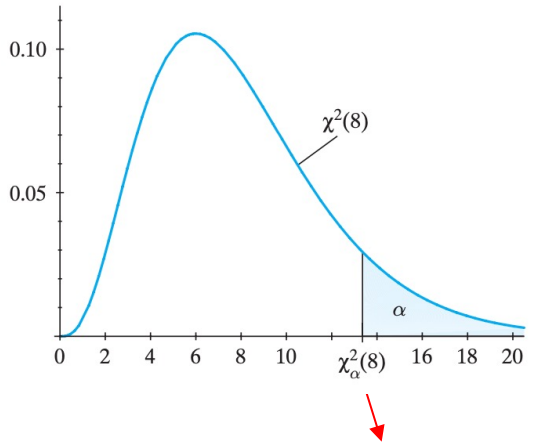
$$\Rightarrow \quad \frac{(n-1)s^2}{\chi^2_{\alpha/2}(n-1)} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}(n-1)}$$

One-sided Upper bound

$$\sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha}(n-1)}$$

One-sided Lower bound

$$\sigma^2 \ge \frac{(n-1)s^2}{\chi^2_{\alpha}(n-1)}$$



chi-squared critical Value

An automatic filling machine is used to fill bottles with liquid detergent. A random sample of 20 bottles results in a sample variance of fill volume of $s^2 = 0.0153$ (fluid ounces)^2. If the variance of fill volume is too large, an unacceptable proportion of bottles will be under- or overfilled. We will assume that the fill volume is approximately normally distributed. A 95% upper-confidence interval is

$$\sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{0.95}(19)}$$

or

$$\sigma^2 \leq \frac{(19)0.0153}{10.117} = 0.0287 \text{ (fluid ounce)}^2$$

This last expression may be converted into a confidence interval on the standard deviation σ by taking the square root of both sides, resulting in

$$\sigma \leq 0.17$$

Therefore, at the 95% level of confidence, the data indicate that the process standard deviation could be as large as 0.17 fluid ounce.



Chi-square Table

	$P(X \leq x)$							
	0.010	0.025	0.050	0.100	0.900	0.950	0.975	0.990
r	$\chi^2_{0.99}(r)$	$\chi^2_{0.975}(r)$	$\chi^2_{0.95}(r)$	$\chi^2_{0.90}(r)$	$\chi^2_{0.10}(r)$	$\chi^2_{0.05}(r)$	$\chi^2_{0.025}(r)$	$\chi^2_{0.01}(r)$
1	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635
2	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.34
4	0.297	0.484	0.711	1.064	7.779	9.488	11.14	13.28
5	0.554	0.831	1.145	1.610	9.236	11.07	12.83	15.09
6	0.872	1.237	1.635	2.204	10.64	12.59	14.45	16.81
7	1.239	1.690	2.167	2.833	12.02	14.07	16.01	18.48
8	1.646	2.180	2.733	3.490	13.36	15.51	17.54	20.09
9	2.088	2.700	3.325	4.168	14.68	16.92	19.02	21.67
10	2.558	3.247	3.940	4.865	15.99	18.31	20.48	23.21
11	3.053	3.816	4.575	5.578	17.28	19.68	21.92	24.72
12	3.571	4.404	5.226	6.304	18.55	21.03	23.34	26.22
13	4.107	5.009	5.892	7.042	19.81	22.36	24.74	27.69
14	4.660	5.629	6.571	7.790	21.06	23.68	26.12	29.14
15	5.229	6.262	7.261	8.547	22.31	25.00	27.49	30.58
16	5.812	6.908	7.962	9.312	23.54	26.30	28.84	32.00
17	6.408	7.564	8.672	10.08	24.77	27.59	30.19	33.41
18	7.015	8.231	9.390	10.86	25.99	28.87	31.53	34.80
19	7.633	8.907	10.12	11.65	27.20	30.14	32.85	36.19
20	8.260	9.591	10.85	12.44	28.41	31.41	34.17	37.57

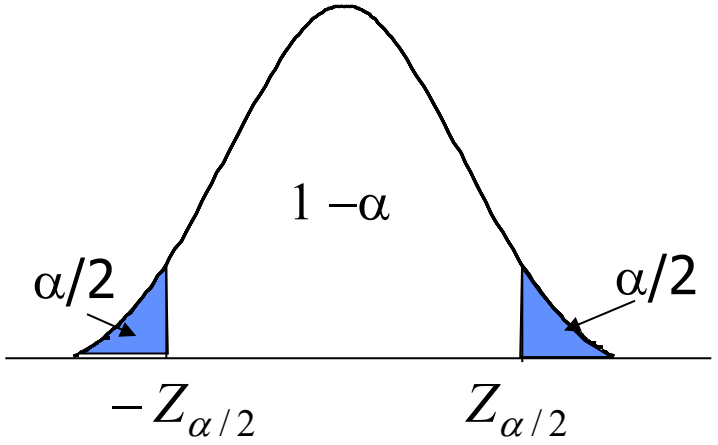
Confidence Interval for Proportion

100(1-α)% CI for a Population Proportion with large sample size

$$X_1, X_2, \cdots, X_n \sim Bernoulli(p)$$

- Using CLT, we can estimate a CI for the population proportion.

$$\hat{p} = \frac{\sum_{i=1}^{n} X_i}{n} \sim N(p, \frac{p(1-p)}{n})$$



$$\hat{p} - Z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \le p \le \hat{p} + Z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

(Approximate) Two-sided CI

$$\hat{p} - Z_{\alpha}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \le p$$

(Approximate) One-sided Lower bound

$$p \le \hat{p} + Z_{\alpha}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

(Approximate) One-sided Upper bound

Example

In a random sample of 85 automobile engine crankshaft bearings, 10 have a surface finish that is rougher than the specifications allow. Therefore, a point estimate of the proportion of bearings in the population that exceeds the roughness specification is $\hat{p} = x/n = 10/85 = 0.12$. A 95% two-sided confidence interval for p is

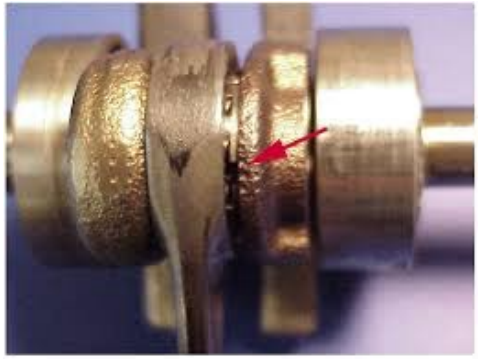
$$\hat{p} - z_{0.025}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \le p \le \hat{p} + z_{0.025}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

or

$$0.12 - 1.96\sqrt{\frac{0.12(0.88)}{85}} \le p \le 0.12 + 1.96\sqrt{\frac{0.12(0.88)}{85}}$$

which simplifies to

$$0.05 \le p \le 0.19$$



Choice of Sample Size

The sample size for a specified value *E* is given by

$$n = \left(\frac{z_{\alpha/2}}{E}\right)^2 p(1-p)$$

An upper bound on *n* is given by (i.e., at least (1-α)100% confidence)

$$n = \left(\frac{z_{\alpha/2}}{E}\right)^2 0.25$$

before

If $\bar{x}$ is used as an estimate of μ, we can be $100(1-\alpha)\%$ confident that the error $|\bar{x} - \mu|$ will not exceed a specified amount E when the sample size is

$$n = \left(\frac{z_{\alpha/2}\sigma}{E}\right)^2$$

Example

Consider the previous example, how large a sample is required if we want to be 95% confident that the error in using $\hat{p}$ to estimate p is less than 0.05?

$$\hat{p} - z_{0.025}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \le p \le \hat{p} + z_{0.025}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$0.12 - 1.96\sqrt{\frac{0.12(0.88)}{85}} \le p \le 0.12 + 1.96\sqrt{\frac{0.12(0.88)}{85}}$$

Using $\hat{p} = 0.12$ as an initial estimate of p

$$n = \left(\frac{z_{0.025}}{E}\right)^2 \hat{p}(1-\hat{p}) = \left(\frac{1.96}{0.05}\right)^2 0.12(0.88) \cong 163$$

OR:

If we wanted to be *at least* 95% confident that our estimate $\hat{p}$ of the true proportion p was within 0.05 regardless of the value of p, we would use Equation 8-27 to find the sample size

$$n = \left(\frac{z_{0.025}}{E}\right)^2 (0.25) = \left(\frac{1.96}{0.05}\right)^2 (0.25) \cong 385$$

Confidence Intervals for Two-Samples

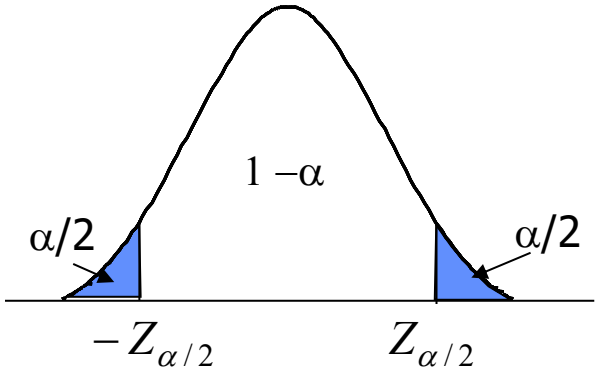
100(1-α)% CI for Differences in Means of Two Normal Distributions (two-sided, known variances)

Two independent random samples from two Normal distributions with the known variances

$$X_1, \ldots, X_n \sim N(\mu_X, \sigma_X^2)$$

$$Y_1, \ldots, Y_m \sim N(\mu_Y, \sigma_Y^2)$$

$$\Rightarrow \bar{X} - \bar{Y} \sim N\left(\mu_X - \mu_Y, \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}\right) \Rightarrow \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}} \sim N(0,1)$$



$$P\left\{-z_{\alpha/2} \leq \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}} \leq z_{\alpha/2}\right\} = 1 - \alpha$$

← **Confidence level**

CI Lower **CI Upper**

$$\bar{X} - \bar{Y} - z_{\alpha/2}\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}} \leq \mu_X - \mu_Y \leq \bar{X} - \bar{Y} + z_{\alpha/2}\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}$$

This CI can be used for mean of non-normal distributions when *n,m*>30

Example

10-4. Two machines are used for filling plastic bottles with a net volume of 16.0 ounces. The fill volume can be assumed normal, with standard deviation $\sigma_1 = 0.020$ and $\sigma_2 = 0.025$ ounces. A member of the quality engineering staff suspects that both machines fill to the same mean net volume, whether or not this volume is 16.0 ounces. A random sample of 10 bottles is taken from the output of each machine.

Machine 1		Machine 2	
16.03	16.01	16.02	16.03
16.04	15.96	15.97	16.04
16.05	15.98	15.96	16.02
16.05	16.02	16.01	16.01
16.02	15.99	15.99	16.00



(b) Calculate a 95% confidence interval on the difference in means. Provide a practical interpretation of this interval.

Example: $\sigma_X = 0.02$, $\sigma_Y = 0.025$

$n = m = 10$

$\bar{X} = 16.015$

$\bar{Y} = 16.005$

$1 - \alpha = 0.95$ $\quad \alpha = 0.05$ $\quad z_{\alpha/2} = 1.96$

$\bar{X} - \bar{Y} = 0.01$

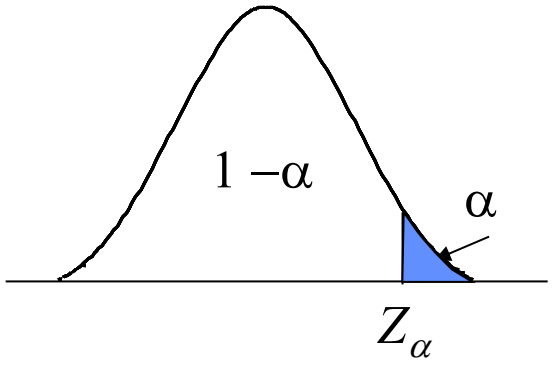
two-sided C.I

$$\bar{X} - \bar{Y} \pm 1.96 \times \sqrt{\frac{\sigma_X^2}{10} + \frac{\sigma_Y^2}{10}}$$

$$0.01 \pm 1.96 \times \sqrt{\frac{0.02^2}{10} + \frac{0.025^2}{10}}$$

$$\Rightarrow [-0.0098, \ 0.0298]$$

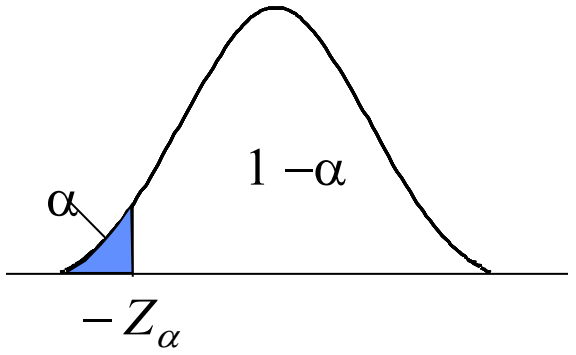
100(1-α)% CI for Differences in Means of Two Normal Distributions (one-sided, known variances)



$$P\left\{\frac{\bar{X}-\bar{Y}-(\mu_X-\mu_Y)}{\sqrt{\frac{\sigma_X^2}{n}+\frac{\sigma_Y^2}{m}}} \le z_\alpha\right\} = 1-\alpha$$

$$\bar{X}-\bar{Y}-z_\alpha\sqrt{\frac{\sigma_X^2}{n}+\frac{\sigma_Y^2}{m}} \le \mu_X-\mu_Y$$

Lower bound



$$P\left\{\frac{\bar{X}-\bar{Y}-(\mu_X-\mu_Y)}{\sqrt{\frac{\sigma_X^2}{n}+\frac{\sigma_Y^2}{m}}} \ge -z_\alpha\right\} = 1-\alpha$$

$$\mu_X-\mu_Y \le \bar{X}-\bar{Y}+z_\alpha\sqrt{\frac{\sigma_X^2}{n}+\frac{\sigma_Y^2}{m}}$$

Upper bound

This CI can be used for mean of non-normal distributions when n,m>30

100(1-α)% CI for Difference in Means of Two Normal Distributions (two-sided, unknown but equal variances)

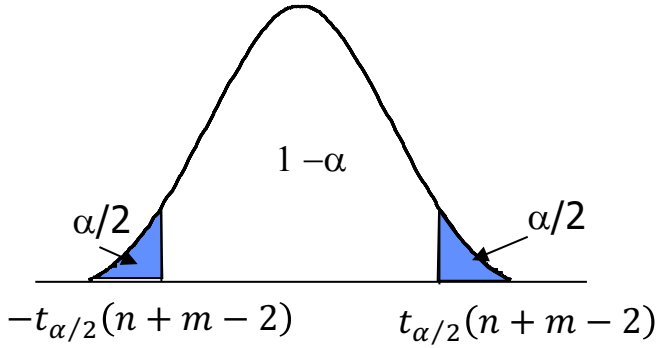
Two independent random samples from two Normal distributions with unknown (but equal) variances

$$X_1, \ldots, X_n \sim N(\mu_X, \sigma^2)$$

$$Y_1, \ldots, Y_m \sim N(\mu_Y, \sigma^2)$$

$$\Rightarrow \quad \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}} \sim t(n + m - 2)$$

$$S_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n + m - 2}$$



$$\Rightarrow \quad P\left\{-t_{\alpha/2}(n+m-2) \le \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}} \le t_{\alpha/2}(n+m-2)\right\} = 1 - \alpha$$

$$\bar{X} - \bar{Y} - t_{\alpha/2}(n+m-2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}} \le \mu_X - \mu_Y \le \bar{X} - \bar{Y} + t_{\alpha/2}(n+m-2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}}$$

Two-sided

100(1-α)% CI for Difference in Means of Two Normal Distributions (one-sided, unknown but equal variances)

Two independent random samples from two Normal distributions with unknown (but equal) variances

$$X_1, \dots, X_n \sim N(\mu_X, \sigma^2)$$
$$Y_1, \dots, Y_m \sim N(\mu_Y, \sigma^2)$$

$$\Rightarrow \quad \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}} \sim t(n+m-2)$$

$$S_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2}$$

$$\Rightarrow \quad \mu_X - \mu_Y \geq \bar{X} - \bar{Y} - t_\alpha(n+m-2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}}$$

Lower bound

$$\mu_X - \mu_Y \leq \bar{X} - \bar{Y} + t_\alpha(n+m-2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}}$$

Upper bound

Example

Example 7.2-2

Suppose that scores on a standardized test in mathematics taken by students from large and small high schools are $N(\mu_X, \sigma^2)$ and $N(\mu_Y, \sigma^2)$, respectively, where σ^2 is unknown. If a random sample of $n = 9$ students from large high schools yielded $\bar{x} = 81.31$, $s_x^2 = 60.76$, and a random sample of $m = 15$ students from small high schools yielded $\bar{y} = 78.61$, $s_y^2 = 48.24$, then the endpoints for a 95% confidence interval for $\mu_X - \mu_Y$ are given by

$$81.31 - 78.61 \pm 2.074\sqrt{\frac{8(60.76) + 14(48.24)}{22}}\sqrt{\frac{1}{9} + \frac{1}{15}}$$

because $t_{0.025}(22) = 2.074$. The 95% confidence interval is $[-3.65, 9.05]$. ■

100(1-α)% CI for Difference in Means of Two Normal Distributions (unknown and unequal variances)

Two independent random samples from two Normal distributions with unknown (and unequal) variances

$$X_1, \ldots, X_n \sim N(\mu_X, \sigma_X^2)$$
$$Y_1, \ldots, Y_m \sim N(\mu_Y, \sigma_Y^2)$$

$$\Rightarrow \quad \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}} \begin{cases} \sim N(0,1) & \text{If } n \text{ and } m \text{ are large enough and the underlying distributions are close to normal} \\ \sim t(r) & \text{for smaller } n \text{ and } m \end{cases}$$

Two-sided **Welch's t-interval**

$$\bar{X} - \bar{Y} - t_{\alpha/2}(r)\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}} \leq \mu_X - \mu_Y \leq \bar{X} - \bar{Y} + t_{\alpha/2}(r)\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}$$

$$r = \left\lfloor \frac{\left(\frac{S_X^2}{n} + \frac{S_Y^2}{m}\right)^2}{\frac{1}{n-1}\left(\frac{S_X^2}{n}\right)^2 + \frac{1}{m-1}\left(\frac{S_Y^2}{m}\right)^2} \right\rfloor$$

$$\mu_X - \mu_Y \geq \bar{X} - \bar{Y} - t_{\alpha}(r)\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}$$

Lower bound

$$\mu_X - \mu_Y \leq \bar{X} - \bar{Y} + t_{\alpha}(r)\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}$$

Upper bound

$\lfloor x \rfloor$ *is the floor function of* x

$$\lfloor 3.4 \rfloor = \lfloor 3.9 \rfloor = \lfloor 3 \rfloor = 3$$

Example

Recall

Example 7.2-2 Suppose that scores on a standardized test in mathematics taken by students from large and small high schools are $N(\mu_X, \sigma^2)$ and $N(\mu_Y, \sigma^2)$, respectively, where σ^2 is unknown. If a random sample of $n = 9$ students from large high schools yielded $\bar{x} = 81.31$, $s_x^2 = 60.76$, and a random sample of $m = 15$ students from small high schools yielded $\bar{y} = 78.61$, $s_y^2 = 48.24$, then the endpoints for a 95% confidence interval for $\mu_X - \mu_Y$ are given by

$$81.31 - 78.61 \pm 2.074\sqrt{\frac{8(60.76) + 14(48.24)}{22}}\sqrt{\frac{1}{9} + \frac{1}{15}}$$

because $t_{0.025}(22) = 2.074$. The 95% confidence interval is $[-3.65, 9.05]$. ■

Q: Now assume $\sigma_X^2 \neq \sigma_Y^2$ and they are both unknown. What is the Welch's t-interval for the difference in mean?

Example: $n = 9$. $m = 15$

$S_X^2 = 60.76$ $S_Y^2 = 48.24$

$$\Rightarrow \frac{\left(\frac{S_X^2}{n} + \frac{S_Y^2}{m}\right)^2}{\frac{1}{n-1}\left(\frac{S_X^2}{n}\right)^2 + \frac{1}{m-1}\left(\frac{S_Y^2}{m}\right)^2}$$

$$= \frac{\left(\frac{60.76}{9} + \frac{48.24}{15}\right)^2}{\frac{1}{8}\cdot\left(\frac{60.76}{9}\right)^2 + \frac{1}{14}\cdot\left(\frac{48.24}{15}\right)^2} = 15.44$$

$\Rightarrow \nu = \lfloor 15.44 \rfloor = 15$, $t_{0.025}(15) = 2.131$

C I.

$$|81.31 - 78.6| \pm 2.131 \times \sqrt{\frac{60.76}{9} + \frac{48.24}{15}}$$

$$= [-4.03, 9.43]$$

100(1-α)% CI for Difference in Means of Two Normal Distributions (paired t-interval)

n independent pairs of (X, Y), but X_i, Y_i for each given *i-th* pair are dependent.

$$X_1, \ldots, X_n \sim N(\mu_X, \sigma_X^2) \qquad Y_1, \ldots, Y_n \sim N(\mu_Y, \sigma_Y^2)$$

$$(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n) \qquad n \text{ independent pairs}$$

$$D_i = X_i - Y_i \qquad i = 1, \ldots, n.$$

$$\Rightarrow \frac{\overline{D} - (\mu_X - \mu_Y)}{S_D/\sqrt{n}} \sim t(n-1) \Rightarrow P\left\{\overline{D} - t_{\alpha/2}(n-1)\frac{S_D}{\sqrt{n}} \le \mu_X - \mu_Y \le \overline{D} + t_{\alpha/2}(n-1)\frac{S_D}{\sqrt{n}}\right\} = 1 - \alpha$$

Two-sided

$$\overline{D} - t_{\alpha/2}(n-1)\frac{S_D}{\sqrt{n}} \le \mu_X - \mu_Y \le \overline{D} + t_{\alpha/2}(n-1)\frac{S_D}{\sqrt{n}}$$

$$\mu_X - \mu_Y \ge \overline{D} - t_{\alpha}(n-1)\frac{S_D}{\sqrt{n}}$$

Lower bound

$$\mu_X - \mu_Y \le \overline{D} + t_{\alpha}(n-1)\frac{S_D}{\sqrt{n}}$$

Upper bound

Example

Example 7.2-4

An experiment was conducted to compare people's reaction times to a red light versus a green light. When signaled with either the red or the green light, the subject was asked to hit a switch to turn off the light. When the switch was hit, a clock was turned off and the reaction time in seconds was recorded. The following results give the reaction times for eight subjects:

Subject	Red (x)	Green (y)	$d = x - y$
1	0.30	0.43	–0.13
2	0.23	0.32	–0.09
3	0.41	0.58	–0.17
4	0.53	0.46	0.07
5	0.24	0.27	–0.03
6	0.36	0.41	–0.05
7	0.38	0.38	0.00
8	0.51	0.61	–0.10

Example: $n=8$, $1-\alpha=0.95$ $\alpha=0.05$ $t_{0.025}(7)=2.365$

$\bar{D}=-0.0625$

$S_d=0.0765$

$$\Rightarrow \quad -0.0625 \pm 2.365 \times \frac{0.0765}{\sqrt{8}}$$

$$[-0.1265,\ 0.0015)$$

100(1-α)% CI for Variances Ratio of Two Normal Distributions

Two independent random samples from two Normal distributions

$$X_1, \dots, X_n \sim N(\mu_X, \sigma_X^2)$$

$$Y_1, \dots, Y_m \sim N(\mu_Y, \sigma_Y^2)$$

$$\Rightarrow \quad \frac{S_X^2 / \sigma_X^2}{S_Y^2 / \sigma_Y^2} \sim F(n-1, m-1)$$

$$F_{1-\alpha}(r_1, r_2) = \frac{1}{F_\alpha(r_2, r_1)}$$

$$\frac{S_X^2}{S_Y^2} F_{1-\alpha/2}(m-1, n-1) \le \frac{\sigma_X^2}{\sigma_Y^2} \le \frac{S_X^2}{S_Y^2} F_{\alpha/2}(m-1, n-1)$$

Two-sided

$$\frac{\sigma_X^2}{\sigma_Y^2} \ge \frac{S_X^2}{S_Y^2} F_{1-\alpha}(m-1, n-1)$$

Lower bound

$$\frac{\sigma_X^2}{\sigma_Y^2} \le \frac{S_X^2}{S_Y^2} F_{\alpha}(m-1, n-1)$$

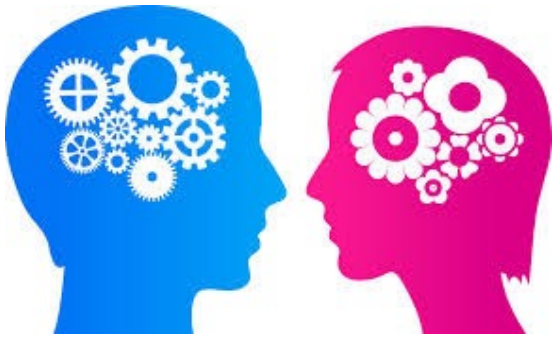
Upper bound

One-sided

Example

10-55. A study was performed to determine whether men and women differ in their repeatability in assembling components on printed circuit boards. Random samples of 25 men and 21 women were selected, and each subject assembled the units. The two sample standard deviations of assembly time were $s_{men} = 0.98$ minutes and $s_{women} = 1.02$ minutes.

(b) Find a 98% confidence interval on the ratio of the two variances. Provide an interpretation of the interval.



Example: $n = 25, \quad m = 21$

$S_X = 0.98 \qquad S_Y = 1.02$

$1 - \alpha = 0.98 \qquad \alpha = 0.02$

$\alpha/2 = 0.01. \qquad 1 - \alpha/2 = 0.99$

$$F_{1-\alpha/2}(20, 24) = F_{0.99}(20, 24) = 0.35$$

$$F_{\alpha/2}(20, 24) = F_{0.01}(20, 24) = 2.74$$

$$\Rightarrow \frac{S_X^2}{S_Y^2} F_{0.99}(20, 24) = \frac{0.98^2}{1.02^2} \times 0.35 = 0.32$$

$$\frac{S_X^2}{S_Y^2} F_{0.01}(20, 24) = \frac{0.98^2}{1.02^2} \times 2.74 = 2.53$$

$$\Rightarrow 0.32 \leq \frac{\sigma_X^2}{\sigma_Y^2} \leq 2.53$$

100(1-α)% CI for Difference in Population Proportions with large sample size

$$\left.\begin{array}{l} X_1, X_2, \cdots, X_{n_1} \sim Bernoulli(p_1) \\ Y_1, Y_2, \cdots, Y_{n_2} \sim Bernoulli(p_2) \end{array}\right\} \rightarrow \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}} \sim N(0,1)$$ For large sample sizes

Two-sided CI

$$(\hat{p}_1 - \hat{p}_2) - Z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} \le (p_1 - p_2) \le (\hat{p}_1 - \hat{p}_2) + Z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

$$(\hat{p}_1 - \hat{p}_2) - Z_{\alpha}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} \le (p_1 - p_2)$$

Lower bound

$$(p_1 - p_2) \le (\hat{p}_1 - \hat{p}_2) + Z_{\alpha}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

Upper bound

One-sided CI

Example

EXAMPLE 10-15 St. John's Wort

Extracts of St. John's Wort are widely used to treat depression. An article in the April 18, 2001, issue of the *Journal of the American Medical Association* ("Effectiveness of St. John's Wort on Major Depression: A Randomized Controlled Trial") compared the efficacy of a standard extract of St. John's Wort with a placebo in 200 outpatients diagnosed with major depression. Patients were randomly assigned to two groups; one group received the St. John's Wort, and the other received the placebo. After eight weeks, 19 of the placebo-treated patients showed improvement, whereas 27 of those treated with St. John's Wort improved.



Fine a 95% confidence interval for the difference of sample proportion of the two groups.

Example. $n = \frac{200}{2} = 100$. $(n_1 = n_2 = n = 100)$

$$\hat{p}_1 = \frac{19}{100} = 0.19$$

$$\hat{p}_2 = \frac{27}{100} = 0.27$$

$1-\alpha = 0.95 \quad \alpha = 0.05 \quad \alpha/2 = 0.025$

$z_{\alpha/2} = 1.96$

95% two-sided C.I

$$\hat{p}_1 - \hat{p}_2 \pm z_{\alpha/2} \cdot \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n} + \frac{\hat{p}_2(1-\hat{p}_2)}{n}}$$

$$= -0.08 \pm 1.96 \times \sqrt{\frac{0.19 \times 0.81}{100} + \frac{0.27 \times 0.73}{100}}$$

$$= -0.08 \pm 0.116$$

$$\Rightarrow \quad [-0.196, \ 0.036]$$

Recap:Summary of Sampling Distributions

		One population	Two populations
Sample Mean	Known Variance	$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$	$\frac{\bar{X}-\bar{Y}-(\mu_X-\mu_Y)}{\sqrt{\frac{\sigma_X^2}{n}+\frac{\sigma_Y^2}{m}}} \sim N(0,1)$
	Unknown Variance	$\frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t(n-1)$	$\frac{\bar{X}-\bar{Y}-(\mu_X-\mu_Y)}{S_p\sqrt{\frac{1}{n}+\frac{1}{m}}} \sim t(n+m-2)$
Sample proportion		$\hat{p} \sim N(p, p(1-p)/n)$	$\frac{(\hat{p}_1-\hat{p}_2)-(p_1-p_2)}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1}+\frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}} \sim N(0,1)$
Sample Variance		$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$	$\frac{S_X^2/\sigma_X^2}{S_Y^2/\sigma_Y^2} \sim F(n-1,m-1)$

Summary

		One population	Two populations
Sample Mean	Known Variance	$\bar{x} - z_{\alpha/2}\sigma/\sqrt{n} \le \mu \le \bar{x} + z_{\alpha/2}\sigma/\sqrt{n}$ $\mu \ge \bar{X} - Z_\alpha \times \frac{\sigma}{\sqrt{n}}$ $\quad$ $\mu \le \bar{X} + Z_\alpha \times \frac{\sigma}{\sqrt{n}}$	$\bar{X} - \bar{Y} - z_{\alpha/2}\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}} \le \mu_X - \mu_Y \le \bar{X} - \bar{Y} + z_{\alpha/2}\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}$ $\bar{X} - \bar{Y} - z_\alpha\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}} \le \mu_X - \mu_Y$ $\quad$ $\mu_X - \mu_Y \le \bar{X} - \bar{Y} + z_\alpha\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}$
	Unknown Variance	$\bar{X} - t_{\alpha/2}(n-1)\left(\frac{S}{\sqrt{n}}\right) \le \mu \le \bar{X} + t_{\alpha/2}(n-1)\left(\frac{S}{\sqrt{n}}\right)$ $\mu \le \bar{x} + t_\alpha(n-1)(s/\sqrt{n})$ $\mu \ge \bar{x} - t_\alpha(n-1)(s/\sqrt{n})$	$\bar{X} - \bar{Y} - t_{\alpha/2}(n+m-2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}} \le \mu_X - \mu_Y \le \bar{X} - \bar{Y} + t_{\alpha/2}(n+m-2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}}$ $\mu_X - \mu_Y \ge \bar{X} - \bar{Y} - t_\alpha(n+m-2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}}$ $\mu_X - \mu_Y \le \bar{X} - \bar{Y} + t_\alpha(n+m-2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}}$
Sample proportion		$\hat{p} - Z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \le p \le \hat{p} + Z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ $\hat{p} - Z_\alpha\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \le p$ $\quad$ $p \le \hat{p} + Z_\alpha\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$	$(\hat{p}_1 - \hat{p}_2) - Z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} \le (p_1 - p_2) \le (\hat{p}_1 - \hat{p}_2) + Z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$ $(\hat{p}_1 - \hat{p}_2) - Z_\alpha\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} \le (p_1 - p_2)$ $\quad$ $(p_1 - p_2) \le (\hat{p}_1 - \hat{p}_2) + Z_\alpha\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$
Sample Variance		$\frac{(n-1)s^2}{\chi^2_{\alpha/2}(n-1)} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}(n-1)}$ $\sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha}(n-1)}$ $\quad$ $\sigma^2 \ge \frac{(n-1)s^2}{\chi^2_\alpha(n-1)}$	$\frac{S_X^2}{S_Y^2}F_{1-\alpha/2}(m-1,n-1) \le \frac{\sigma_X^2}{\sigma_Y^2} \le \frac{S_X^2}{S_Y^2}F_{\alpha/2}(m-1,n-1)$ $\frac{\sigma_X^2}{\sigma_Y^2} \ge \frac{S_X^2}{S_Y^2}F_{1-\alpha}(m-1,n-1)$ $\quad$ $\frac{\sigma_X^2}{\sigma_Y^2} \le \frac{S_X^2}{S_Y^2}F_\alpha(m-1,n-1)$

Summary

- **Find interval estimate for parameters (rather than point estimator).**
- **Concepts: confidence level, two-sided, one-side confidence interval**

$$\mathrm{P}(\theta \in [L, U]) = 1 - \alpha$$

- **How to determine the minimum number of samples needed to achieve certain confidence width (i.e., error).**